

Calculus I Lesson 16B

1) $f(x) = x^2 + 3x + 5$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}} ?$$

2) $f(x) = 4x^2 - 7x + 1$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}} ?$$

3) $f(x) = x^4 + 3x^2 + 10x - 1$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}} ?$$

$$4) f(x) = 4x^{-2} + 13x^2 + 5x - 11$$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}}?$$

$$5) f(x) = 5x^{-4} + 13x^{-3} + 5x^{-2} - 4$$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}}?$$

$$6) f(x) = x^{1/2} + 3x^{1/3} + 5x^{1/4} - 7$$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}}?$$

$$7) f(x) = \frac{4x^3 + 3x + 5}{x^2}$$

$$\text{Hint: } \frac{4x^3 + 3x + 5}{x^2} = \frac{4x^3}{x^2} + \frac{3x}{x^2} + \frac{5}{x^2} = 4x + \frac{3}{x} + \frac{5}{x^2} = 4x + 3x^{-1} + 5x^{-2}$$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}}?$$

$$8) f(x) = \frac{2x^4 + 7x^2 + 15}{x^3}$$

$$\text{Hint: } \frac{2x^4 + 7x^2 + 15}{x^3} = \frac{2x^4}{x^3} + \frac{7x^2}{x^3} + \frac{15}{x^3} = 2x + \frac{7}{x} + \frac{15}{x^3} = 2x + 7x^{-1} + 15x^{-3}$$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}} ?$$

$$9) f(x) = \frac{x^3 + 2x + 1}{x^{1/2}}$$

$$\begin{aligned}\text{Hint: } \frac{x^3 + 2x + 1}{x^{1/2}} &= \frac{x^3}{x^{1/2}} + \frac{2x}{x^{1/2}} + \frac{1}{x^{1/2}} \\ &= x^{3-1/2} + 2x^{1-1/2} + x^{-1/2} \\ &= x^{5/2} + 2x^{1/2} + x^{-1/2}\end{aligned}$$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}} ?$$

